SPE-227817-MS

Advanced MPD Techniques for Ballooning Control: Effective Management of Wellbore Breathing and Overbalance Optimization Allowed Successful Navigation Through the 12 in. Section to Achieve Target Depth – Case Study

S. E. Krekov, Weatherford; M. Herhar, Weatherford, Saudi Arabia

 DOI 10.2118/227817-MS

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

This study explores advanced managed pressure drilling (MPD) techniques to address well-control challenges caused by wellbore ballooning, compounded by high formation-charged mud volumes and aggressive mud contamination. The goal is to demonstrate successful unconventional MPD applications for depleting charged mud volumes during conventional well control operations. The approach aims to mitigate formation breathing effects and reduce overbalance pressure on weak formations to successfully drill the 12 in. section to the planned total depth (TD).

Due to aggressive flow behavior and mud weight reduction from saltwater influx, a methodical approach was implemented to reduce charged volumes in the well through controlled reductions in bottomhole equivalent circulating density (BH ECD). Each pressure decrement stabilized the well and returned charged volumes to the wellbore. Maintaining pressure balance was crucial, with steps executed under controlled conditions followed by circulation in standpipe pressure (SPP) control mode to keep constant bottomhole pressure (CBHP) despite non-homogeneous annular fluids.

The ballooning release technique required precise execution to ensure well stability. Critical requirements included establishing baseline parameters before circulation initiation. These parameters were references for stable conditions, enabling accurate monitoring and adjustments. Given the well's pressure sensitivity, SPP control during influx circulation was essential.

Successful execution of all decrement steps combined with SPP-controlled circulation reduced BH ECD from 150.2 to 146.7 pcf. This reduction decreased overbalance on the low-pressure formation, mitigating the risk of severe or total losses. The well was drilled to TD through careful parameter management while maintaining acceptable loss rates. This methodical approach preserved wellbore stability, minimized operational risks, and ensured optimal well control throughout drilling operations.

Introduction

Modern drilling increasingly encounters wells with complex geological profiles and narrow drilling windows, further complicated by formation breathing effects caused by microfractures. This case study demonstrates that traditional well control methods often prove ineffective in such scenarios. Conventional drilling with standard mud weights introduces additional risk due to increased overbalance during dynamic conditions with annular friction losses when the well is already statically overbalanced. These technical challenges underscore the critical importance of managed pressure techniques during circulation, a principle well-documented in the technical literature, including AADE-16-FTCE-53 from the American Association of Drilling Engineers (Ian L Everhard, et al. 2016).

Wellbore ballooning, known as formation breathing, occurs in wells where subsurface formation characteristics, including lithology, geomechanical integrity, pore pressure distribution, and fracture pressure gradients, are not fully defined. This phenomenon presents significant diagnostic and operational challenges, particularly in formations containing natural fractures.

The ballooning effect manifests during pressure fluctuations, particularly mud pump cycling. Equivalent circulating density (ECD) increases during circulation due to combined hydrostatic and annular frictional pressure components. When dynamic pressure marginally exceeds local fracture initiation thresholds, drilling fluid enters natural microfractures. ECD returns to static mud weight conditions upon pump cessation, often insufficient to maintain open fractures. As microfractures close, intruded fluid returns to the wellbore, observable as surface mud volume gains during non-circulation periods. Hydrocarbon traces in returning fluid may further complicate diagnosis.

Figure 1 illustrates the standard wellbore ballooning (formation breathing) process. When the pumps are turned off, the status changes, leading to the closure of the fracture and allowing fluid to flow back into the wellbore. Once the pumps are turned on, the fracture reopens, and the circulating drilling fluid begins to enter and pressurize the formation fractures.

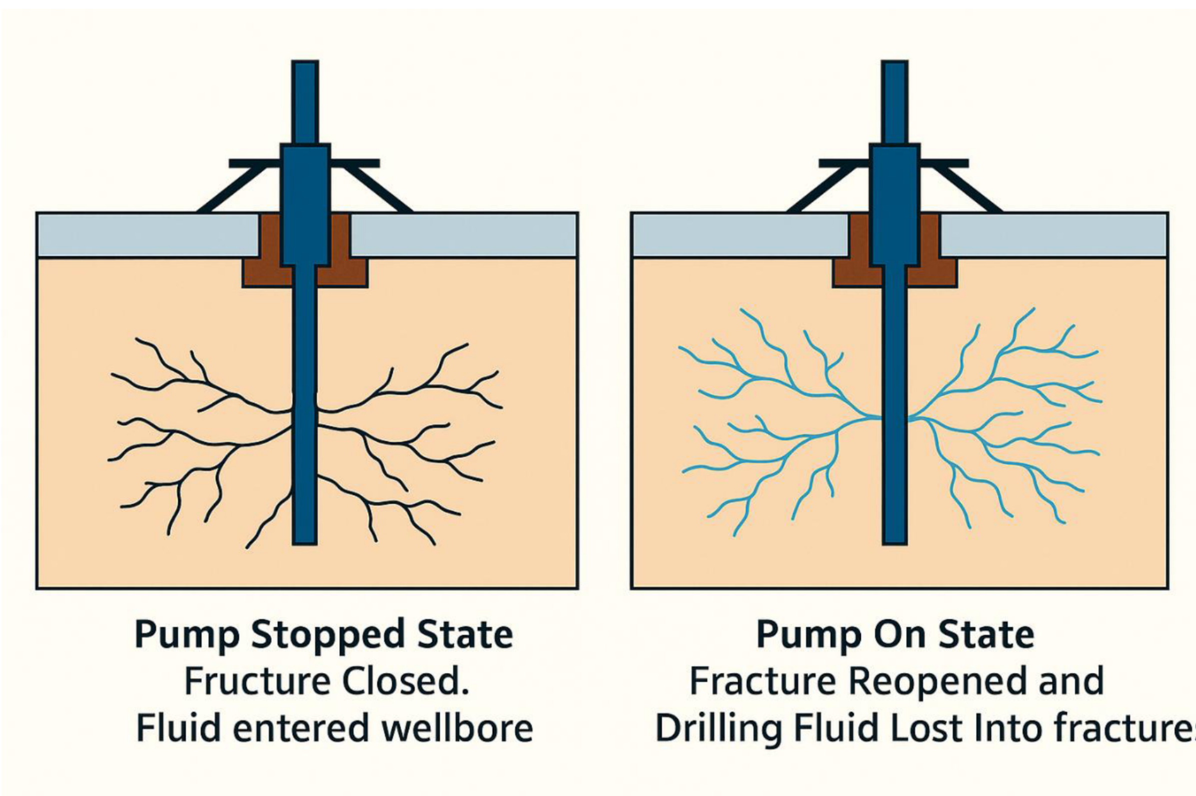

Figure 1—Wellbore Ballooning illustration.

As documented in SPE-216933-MS (Rodrigo Varela and Mohammed Julaidan et al. 2023), such losses sometimes remain undetected. In that case, ballooning can be initially misinterpreted as a gain/loss event, especially when mud losses remain undetected. A slight flow-out increase prompted the well shut-in. Misdiagnosis led to unnecessary well control procedures, resulting in 15 days of non-productive time (NPT).

Flowback events are frequently mistaken for formation influxes (kicks), triggering well control procedures, including shut-in and kill operations. When ballooning is incorrectly treated as a well control event, these procedures may escalate situations unnecessarily, potentially causing increased NPT, wellbore instability, or even well abandonment.

Resolution attempts may require weeks or months to achieve a well-equilibrium state, as mud weighting and circulation are time-intensive processes. This often creates cycles of mud weighting followed by loss control efforts. Even after extensive work, subsequent operations face challenges due to persistent well sensitivity, low loss-cure probability, and worsening losses from increased mud weights.

Given conventional methods' limitations, adopting alternative approaches becomes imperative to enhance operational safety and minimize NPT. Managed Pressure Drilling (MPD) represents one of the most effective solutions, enabling precise bottomhole ECD adjustments.

The international association of drilling contractors (IADC) defines MPD as an adaptive drilling process that precisely controls the annular pressure profile throughout the wellbore. Its objectives include determining downhole pressure environmental limits and managing the annular hydraulic pressure profile accordingly (ABS Guide for Classification and Certification of Managed Pressure Drilling Systems, September 2017).

MPD offers substantial advantages over conventional methods, particularly in complex or narrow-margin wells. Benefits include:

- Reduced ballooning misinterpretation
- Improved drilling in depleted zones
- Enhanced wellbore stability in unconsolidated formations
- Lower formation damage risk
- Real-time monitoring adaptability

MPD also enhances pressure control, improves kick/loss detection, reduces NPT, increases drilling rates, and improves environmental safety and cost efficiency.

During MPD operations, drilling through highly charged formations makes this effect pronounced not only during pump cycling but also dynamically as pressure deviates from balance points or under loss conditions. Minor pressure reductions may cause noticeable return flow increases. The data graphing in Figure 2 demonstrates a clear example of this statement. As can be shown, an approximately 20 psi reduction in SBP resulted in a noticeable increase in return flow. The flow trend can be noticed to decrease over time. However, when SBP was returned to its original value, stabilized losses were observed to respond to that increase.

Nevertheless, the identification of ballooning in the MPD technique may differ slightly from conventional methods due to precise Coriolis flowmeter measurements and real-time flow-out trend monitoring. Beyond traditional Horner plots, MPD provides a comprehensive understanding of wellbore behavior. A key advantage of MPD lies in its real-time adaptability, which facilitates early kick detection, improves wellbore stability, and minimizes formation damage. This approach supports effective management of high-pressure environments, enhances data-driven decision-making through continuous visual monitoring, reduces drilling fluid consumption, and improves overall operational efficiency.

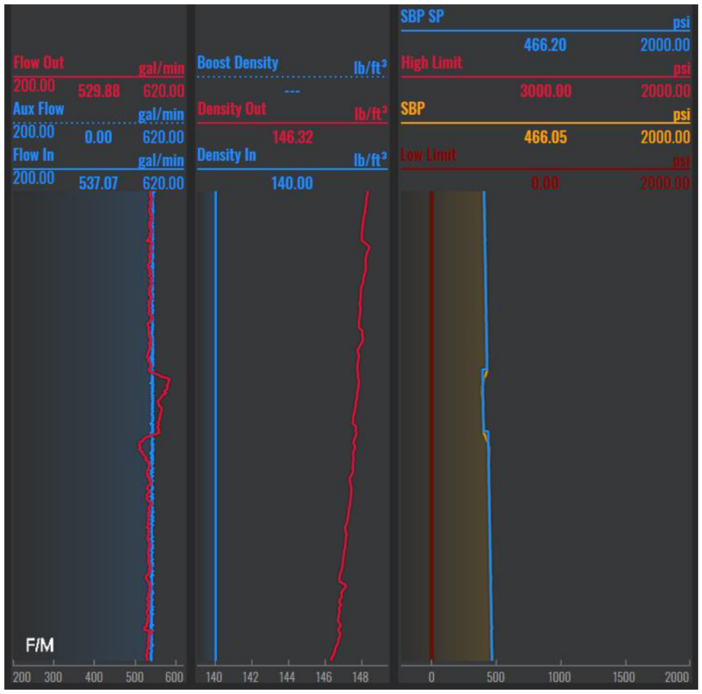

Figure 2—Ballooning in Dynamic conditions.

Formation Characteristics

The target formation in this well consisted primarily of carbonates. Drilling proceeded through a high-pressure zone where a kick was encountered during conventional operations before reaching the low-pressure target zone. This led to a kick/loss scenario before entering these zones.

Low-pressure zone drilling presents significant risks, particularly regarding induced fractures and mud losses. Operations grow more complex in deeper low-pressure formations, especially when encountering overlying high-pressure water-bearing zones with low permeability. These Permian-period compacted, low-pressure formations typically comprise carbonates such as limestone, dolomite, or dolomitic limestone, often interbedded with anhydrite streaks and shale layers.

Offset data indicate that fracture pressures within these zones typically range between 142 pcf and 155 pcf, while the pore pressures of the overlying formations may reach values between 132 and 145 pcf. This narrow margin creates significant operational challenges, compounded by induced fractures and formation breathing. Therefore, advanced pressure management and well control strategies are essential to ensure safe and effective operations.

Drilling Challenges

The operation was initially performed utilizing conventional drilling techniques with a rotating control device (RCD) installed for enhanced safety. The RCD enables pipe rotation during drilling while diverting flow from the rig floor. The RCD also enables pipe rotation should a shut-in be required, compared to a stationary period if the well is shut-in at the rig's BOP. This is limited by the RCD maximum allowable applied surface pressure (MAASP).

A kick was encountered shortly after drilling out from the previous casing point. It was initially controlled using 147 pcf BH ECD before developing into a kick/loss scenario, complicating pressure management and well control.

Fracture initiation occurred just below the casing shoe during the well control event. After observing a flow increase, the well was shut in with 138 pcf mud weight. Control was regained at 144 pcf, but secondary influxes required further increases to 147 pcf, creating a week-long influx and loss mitigation cycle. MPD deployment ultimately restored the wellbore stabilization and integrity and prevented further instability.

Figure 3 Shows the planned 12 in. section profile.

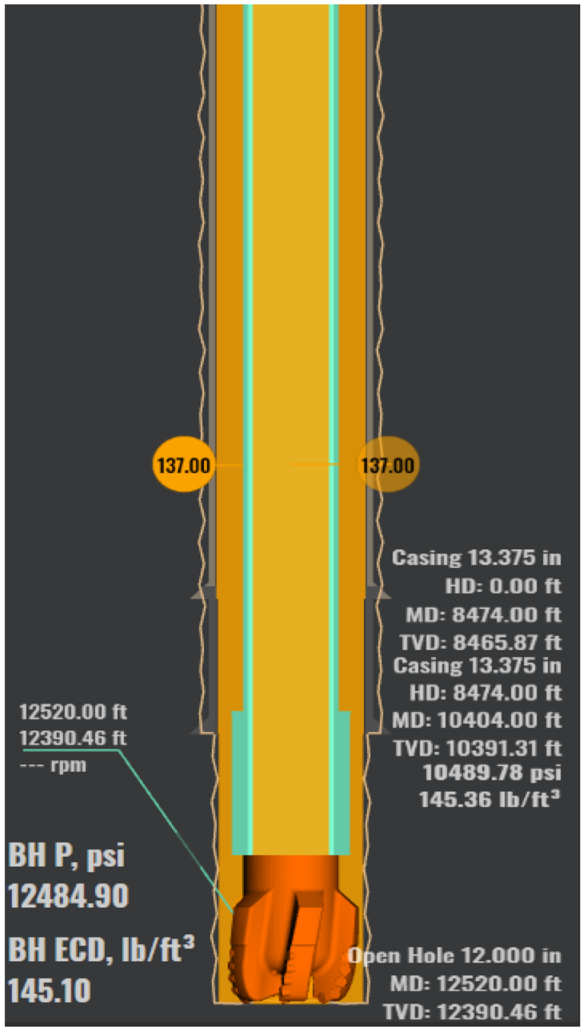

Figure 3—Wellbore Profile.

Kick/loss situations present unique challenges due to simultaneous formation fluid influx and drilling fluid loss. This dual imbalance creates complex well control scenarios because:

1. **Conflicting Pressure Management Requirements:** Kick prevention requires bottomhole pressure (BHP) exceeding formation pressure, while loss prevention requires BHP below fracture pressure. Balancing between these limits becomes extremely difficult, often creating limited to no drilling windows.
2. **Well Control Risks:** Introducing formation fluids (typically gas or water) during influx events can rapidly escalate into severe well control incidents. Simultaneously, fluid losses diminish hydrostatic pressure, creating conditions conducive to further influxes. When these phenomena occur concurrently, they may precipitate underground blowouts, particularly hazardous scenarios that present extreme operational challenges for containment and control.
3. **Dynamic Interactions:** Losses reduce hydrostatic pressure, potentially triggering kicks, while kicks may increase annular pressure, worsening losses by fracturing the formation.

4. **Fluid Uncertainty:** Both situations introduce unknowns, formation fluids from kicks and variable loss zones, complicating pressure and flow calculations.
5. **Equipment and Monitoring Complexity**: Detecting and responding to such events is complex due to the rig system sensors' lack of resolution for simultaneous kick/loss detection.
6. **Operational Impacts:** Managing these events safely often requires drilling pauses, well control operations, or sidetracking, increasing costs and time.

Each scenario presents unique operational challenges where MPD delivers substantial value. This value manifests both proactively, preventing issues during well planning and design, and reactively, effectively addressing challenges that emerge during operations. This paper examines two case studies demonstrating MPD's dual capability: as a preventive engineering solution and as a responsive mitigation technique.

Scenario 1: Proactive MPD Implementation

Early MPD adoption effectively mitigates complex wellbore conditions like ballooning, particularly in narrow-margin formations with difficult-to-detect losses. Key advantages of MPD include early loss detection, reduced initial overbalance, and accurate pore pressure determination during drilling.

Accurately identifying and quantifying fluid losses during conventional drilling operations remains a significant challenge in many regions. Conventional operations often struggle with loss identification, particularly when masked by simultaneous rig activities such as mud transfers or adding water and chemical treatments to the active system. For example, continuous 10 bbl/hr water additions may obscure formation losses undetectable by standard sensors.

Industry publications consistently demonstrate MPD's superior value compared to conventional operations. A particularly instructive case in SPE-216933-MS (Rodrigo Varela et al. 2023) documents how misinterpreting formation breathing as a fluid influx led to an unjustified mud weight increase from 80 to 109 pcf. This adjustment created a dangerous overbalance situation (29 pcf) above the original weight and a staggering (41 pcf) above actual formation pressure.

The excessive overbalance severely compromised well integrity, producing unreliable cement bond logs (CBL) and ultrasonic evaluations due to poor tool response. Tool eccentricity worsened these inaccuracies, as gravitational forces forced the logging assembly against the wellbore's low side, introducing measurement bias. The resulting data deficiencies hindered definitive cement assessment, creating uncertainty in well integrity verification and impacting long-term operational decisions.

A cement plug was placed without verifying complete displacement to address the suspected wellbore breathing, followed by a negative pressure test with a packer. The test confirmed a pore pressure below 68 pcf, proving the mud weight had been unnecessarily overestimated. This demonstrated that the well could have reached TD safely without the excessive fluid densities previously employed (Quote taken from Rodrigo Varela and Mohammed Julaidan, SLB, Dammam, Kingdom of Saudi Arabia. et al. 2023. Analysis of the Ballooning Effect: No Loss, Only Gain).

This case highlights a critical limitation of conventional well control methods: the tendency to unnecessarily escalate mud weights in response to perceived influxes, which risks formation damage and complicates later operations. Proactive MPD could have mitigated these issues through real-time annular pressure profile control, precise diagnostic techniques like SBP modulation for pore pressure verification and eliminating repetitive fluid displacement cycles to condition the drilling fluid, homogenizing its properties.

In summary, proactive MPD significantly enhances both operational efficiency and wellbore integrity through multiple mechanisms. By enabling early detection of fluid losses and providing precise monitoring of return flow rates, MPD allows for more accurate maintenance of wellbore pressure balance.

This capability prevents unnecessary mud weight increases while simultaneously facilitating real-time estimation of formation pressure. When these technical advantages are properly integrated during the well planning phase, they collectively minimize risks associated with wellbore ballooning and stability issues, ultimately leading to safer and more cost-effective drilling operations.

Scenario 2: Reactive Application of MPD Following an Incident and Effective Management of Formation Breathing to Achieve Section TD

This scenario investigates MPD's use in addressing well control issues, particularly those arising from wellbore ballooning exacerbated by highly pressurized formation mud volumes and significant contamination, as detailed below.

Case Study

A comprehensive evaluation of the implemented procedures requires systematically analyzing initial well conditions and chronological operational developments. While preceding sections have detailed the formation's geological and pressure challenges, this analysis explicitly examines the operational timeline from initial conventional well control attempts to the eventual deployment of MPD solutions.

The 12 in. hole section drilling was initiated using an optimized mud weight of 138 pcf, established through field experience. At approximately 50 feet below the casing shoe, drilling operations encountered the first influx, identified through pit volume totalizer (PVT) anomalies and confirmed by positive flow checks. The well was immediately shut in, recording shut-in pressures of 590 psi (SICP) and 436 psi (SIDP), indicating a formation pressure gradient of approximately 144 pcf.

Following an unsuccessful attempt to resume drilling, a secondary well control operation was performed to stabilize the well using 147 pcf kill-weight mud. Subsequent circulation revealed significant fluid contamination, with the return mud weight dropping to 130 pcf and multiple indicators of formation fluid influx, such as a surge in chloride content from 12,000 to 32,000 ppm, a decrease in electrical stability (E.S.) from 470 V to 220 V, a decline in the oil-water ratio to 74/26, and noticeable increases in both plastic viscosity (PV) and yield point (YP).

Despite achieving apparent well control after the kill operation, attempts to restore circulation encountered severe losses of 140 bbl/hr at 300 gpm, preventing the planned 600 gpm flow rate. Conventional stabilization efforts proved ineffective, prompting suspicion of wellbore ballooning as a key factor. The well's behavior suggested initial undetected fluid losses through natural microfractures, followed by partial returns when dynamic pressure decreased, characteristic ballooning phenomena. This cyclic gain-loss pattern indicated three probable flow zones:

1. Naturally fractured formations exhibiting classic ballooning behavior,
2. Induced fracture networks from previous well control actions, or
3. The primary target formation's inherent loss-prone characteristics.

For six days, conventional well control methods struggled to stabilize the well. With an RCD already in place, operations transitioned to MPD methodology. The MPD implementation commenced with a carefully controlled mud weight reduction from 148 pcf to 140 pcf, during which formation fluid losses became evident. The well demonstrated remarkable pressure sensitivity, strongly indicating active ballooning behavior across multiple fracture networks.

While ballooning typically manifests during pump cycles, in this instance, it became apparent during displacement as the lighter MPD mud reduced the BHP. The well's acute response to minor pressure variations confirmed this, which persisted even after controlling the fluid gain with increased ECD.

This case study underscores the inherent constraints of conventional well control when addressing complex wellbore dynamics, while highlighting MPD's advantages in precise pressure management, loss prevention, and early formation response detection, all achieved without requiring multiple mud weight adjustments.

The formation's exceptional sensitivity was evidenced by measurable responses to BHP fluctuations as small as 10 psi. Post-circulation analysis revealed substantial mud contamination, with density dropping sharply from 140 pcf to 136 pcf, accompanied by significant volumes of contaminated fluid. The accompanying figures (Figure 4 and Figure 5) present representative operational data documenting these observed phenomena.

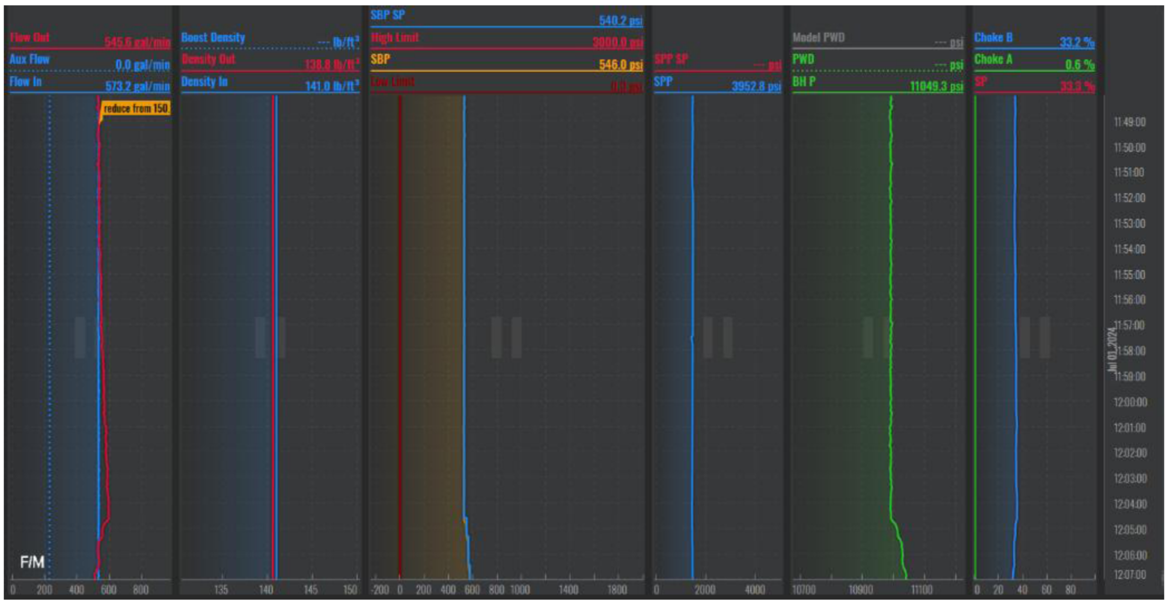

Figure 4—Formation ballooning response to slight surface pressure changes.

Figure 5—Formation fluids at surface post-circulating bottom-up.

Ballooning Bleed-off Test

After several days of drilling with minimal fluid losses (20-30 bbl/hr), a ballooning bleed-off test was conducted during the initial 12 in. section. Weatherford initiated a Management of Change (MOC) process to temporarily increase the MPD operational matrix limits from 5 bbl to 30 bbl for the test duration. Before testing, comprehensive baseline parameter fingerprinting was performed to establish monitoring references, accounting for saltwater contamination effects on mud properties.

Procedures:

Below is the pre-test developed Procedure for the MPD ballooning bleed-off test:

1. Wellbore Preparation:
   - ○ Circulate to clean the hole and condition the drilling fluid until mud weight in (MW IN) equals mud weight out (MW OUT).
2. Baseline Establishment:
   - Record initial parameters for:
     a. Flow rate in (Flow IN)
     b. Flow rate out (Flow OUT)
     c. Input fluid density (Density IN)
     d. Return fluid density (Density OUT)
     e. Standpipe pressure (SPP)
     f. MPD surface backpressure (SBP)
3. **Initial Pressure Reduction:**
   - Reduce equivalent mud weight (EMW) by 0.2 pcf increment.
   - Monitor flow-out trend, which should temporarily increase before stabilizing and gradually decreasing (characteristic ballooning response).
4. **Circulation Monitoring:**
   - Maintain constant standpipe pressure during bottom-up circulation.
   - Observe SPP trends, noting that the MPD choke automatically adjusts SBP to compensate for annular fluid contamination.
5. **Stabilization Criteria:**
   - Continue monitoring until:
     - MW OUT stabilizes at MW IN
     - Flow trends show consistent stabilization
   - If parameters remain stable, proceed with additional 0.2 pcf reduction stages until the target EMW is 147 pcf.

The ballooning bleed-off test is a technically sophisticated operation that requires meticulous execution, particularly in high-volume charged formations where differentiating between ballooning effects and genuine influxes presents significant diagnostic challenges. The procedure requires comprehensive parameter monitoring beyond standard protocols, beginning with precise fingerprinting of baseline conditions.

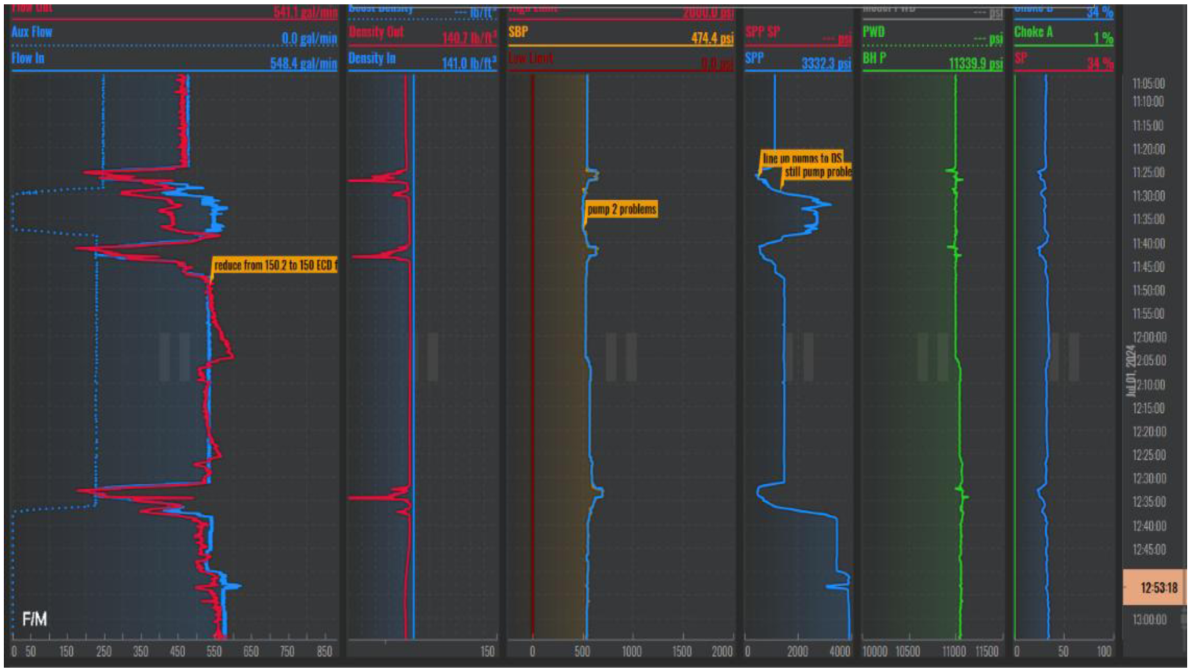

Figure 6—MPD ballooning bleed-off test.

Verifying mud weight equilibrium (MW In = MW Out) is critical to this process, as return fluid density is a primary indicator of formation fluid contamination. Equally vital is standpipe pressure (SPP) management, which maintains constant BHP, an especially crucial consideration when handling saltwater-contaminated annular fluids that could otherwise disrupt pressure balance. These measured parameters collectively enable operators to distinguish between normal ballooning behavior and actual well control events during the sensitive pressure reduction process.

A thorough understanding of Surface Back Pressure (SBP) is essential for defining operational boundaries and establishing safe initial conditions. Two principal methodologies ensure secure circulation:

1. **SPP-Controlled Mode:** Maintaining active standpipe pressure regulation.
2. **Manual SBP Adjustment:** Holding SPP constant while manually modulating backpressure.

Both techniques demand exacting process control and continuous monitoring to distinguish between formation ballooning and actual influx events. The operation was executed through six precisely controlled phases, implementing a staged reduction of ECD in decrements ranging from 0.2 to 1.1 pcf per phase. This systematic pressure decrease reduced bottomhole ECD from 150.2 pcf to 146.7 pcf.

Following each pressure reduction, standard procedures were implemented, including circulation in SPP control mode to maintain stable BHP, removal of contaminated drilling fluid, thorough mud conditioning, and complete homogenization of both incoming and return mud systems to ensure consistent fluid properties before proceeding to subsequent stages.

While post-circulation analysis showed no gas influx, elevated LEL readings were observed. The operation generated over 800 barrels of saltwater-contaminated drilling fluid.

Implementing these measures successfully mitigated excessive overbalance pressure on the vulnerable formation, permitting drilling operations to continue despite the ineffectiveness of conventional lost circulation material (LCM) treatments. By strategically applying MPD methodologies, operators could safely navigate the complex wellbore conditions, achieving successful well construction to the planned TD.

Time	BH ECD (pcf)		CSG Shoe ECD (pcf)	SBP (psi)	MPD Flow		Mud Pumps FR (GPM)	SPP (psi)	MW (pcf)		Gained Volume (bbls)	Time to stabilize (minutes)	total contamination mud isolated
	from	*to*			*in*	*out*			in	out cut			
16:30	150	149.8	149.8	480	540	567	555	3380	140+	137	30	44	200
19:00	149.8	149.3	149.3	522	534	533	555	3360	140+	138	27	33	0
1:00	147.4	147.2	147.2	311	534	524	555	3420	141	140+	10	30	0
7:00	147.5	147.8	147.8	377	535	535	555	3390	140+	137	14	13	250
10:00	147.8	146.7	146.8	335	523	520	555	3340	140+	135+	20	30	300

Updated MPD system with actual mud rheology

Figure 7—MPD ballooning bleed-off test results.

Real results achieved

The systematic execution of controlled pressure decrements, combined with precise standpipe pressure (SPP)-managed circulation, proved instrumental in significantly reducing bottomhole equivalent circulating density (BH-ECD) from 150.2 pcf to 146.7 pcf. This carefully engineered pressure decrease successfully mitigated overbalance conditions across the low-pressure formation zone, substantially decreasing risks of both formation impairment and fluid loss incidents.

Drilling crews reached TD by maintaining rigorous parameter control throughout operations while sustaining loss rates within acceptable operational thresholds. The methodical implementation of this approach delivered dual benefits: preserving optimal wellbore stability while simultaneously reducing drilling hazards, resulting in consistently maintained well control integrity from initial penetration through final TD achievement.

Conclusion

This study demonstrates that:

1. MPD enables precise formation breathing management, distinguishing ballooning from kicks
2. Controlled ECD reduction mitigates overbalance and minimizes losses in narrow-margin wells.
3. Proactive MPD implementation prevents ballooning misdiagnosis and reduces NPT.

Recommendations

1. Integrate MPD during planning for suspected ballooning formations.
2. Employ Coriolis flowmeters for enhanced diagnostics.
3. Standardize ballooning tests with staged ECD reductions.

Acknowledgements

The authors wish to thank engineering management, operations teams, wellsite crews and our service partners for their contribution and support in publishing this paper.

Nomenclature

MPD	*Managed Pressure Drilling*
TD	*Total Depth.*
BHP	*Bottomhole Pressure.*
CBHP	*Constant Bottomhole Pressure.*
ECD	*Equivalent Circulating Density.*
BH ECD	*Bottom Hole Equivalent Circulating Density.*
SPP	*Standpipe Pressure.*
LCM	*Lost Circulation Material.*
NPT	*Non-Productive Time.*

IADC	*International Association of Drilling Contractors.*
SBP	*Surface Back Pressure.*
BOP	*Blowout Preventer*
MAASP	*Maximum Allowable Annular Surface Pressure.*
RCD	*Rotating Control Device.*
SICP	*Shut-In Casing Pressure.*
SIDPP	*Shut-In Drillpipe Pressure.*
PV	*Plastic Viscosity.*
YP	*Yield Point.*
MOC	*Management Of Change.*
MW	*Mud Weight.*
EMW	*Equivalent Mud Weight.*
LEL	*Lower Explosive Limit.*

References

Ian L Everhard, Juan Espinoza, Iain Maley, Baker Hughes; Alejandro Montiel, PEMEX, et al 2016. Drilling on the Coast of Tabasco Mexico: Battling Wellbore Instability in a Narrow Pressure Window. This paper was prepared for presentation at the 2016 AADE Fluids Technical Conference and Exhibition held at the Hilton Houston North Hotel, Houston, Texas, April 12-13, 2016. AADE-16-FTCE-53.

Rodrigo Varela and Mohammed Julaidan, SLB, Dammam, Kingdom of Saudi Arabia. et al 2023. Analysis of the Ballooning Effect: No Loss, Only Gain. This paper was prepared for presentation at the *ADIPEC* held in Abu Dhabi, UAE, 2–5 October, 2023. SPE-216933-MS. https://doi.org/10.2118/216933-MS.

ABS Guide for Classification and Certification of Managed Pressure Drilling Systems, September 2017. Managed Pressure Drilling (MPD) | Oil and Gas Drilling Glossary | IADCLexicon.org